



AI-Driven Information Systems Innovation

Overview

The goal of this group project is to apply the concepts, frameworks, and analytical tools learned in class to analyze a company's **core** management information systems (MIS) and its role in supporting business strategy and value creation. Building on this analysis, the project requires proposing an **AI-driven solution** to transform and enhance the selected MIS.

Through this project, you will connect theory to practice by integrating strategic and technological perspectives, demonstrating how AI can transform MIS to create business value and competitive advantage.

Group Formation: Groups will be **randomly assigned** in class. All deliverables are group-based. Separate assignments will be created in Canvas for each stage, and each group is required to submit **only one copy per stage** (i.e., proposal, presentation slides, final report).

Project Timeline

Stage	Deliverables	Deadline
Proposal	6–8 pages, 12pt, 1.5-line spacing	May 9 (Sat), 23:59
Presentation	20 min presentation + 5 min Q&A	May 17 (Sun), 09:00
Final Report	~15 pages, 12pt, 1.5-line spacing	May 23 (Sat), 23:59

Stage 1 – Proposal

Deadline: May 9 (Saturday), 23:59 (Beijing Time)

Deliverable: A Word document (6–8 pages, Times New Roman, 12-point font, 1.5-line spacing) with page numbers, group information, a clear title, and well-organized sections.

Goal: Conduct a strategic and information system analysis of a real company using the frameworks and analytical tools learned in class, and identify a focal (not more than three) information-system problem(s) that the Stage 2 AI solution will address.

Company Selection: Select a company you are familiar with or interested in. Companies where you have work or internship experience are recommended, as first-hand familiarity will sharpen your problem diagnosis in Stage 1 and the realism of your solution in Stage 2

Scope and Focus: Most companies operate multiple products and information systems, so a clear focus is required. The strategic analysis should center on the company's primary product or service, while the information systems analysis should focus on **one core system** that supports key value-generating activities. For example, instead of broadly analyzing "Tencent's MIS," narrow the scope to a specific product and system, such as WeChat or QQ Music.

Data Sources: Use credible sources to support your analysis, such as company reports, public filings, academic articles, and reliable online materials. Do not use the fabricated statistics generated by AI.

Proposal Structure (suggested):

- 1) Introduction: Brief overview of the company and its core business.
- 2) Strategic Analysis: Apply relevant frameworks (e.g., *Five-Force*, *Value Chain*, *Resource-Based View*, *IS Strategy Triangle*) to analyze the competitive environment, key activities, and strategic resources.
- 3) Information System Analysis: Identify one core MIS. Explain its main functions, purpose, and role in supporting the company's strategy, and evaluate its effectiveness in creating business value.
- 4) Focal Problem(s) (initial articulation): Identify one specific, information-system-related problem within the system you analyzed — not a general management or corporate issue. Begin applying the Problem Articulation Template introduced in Stage 2 (Actor, Context, Symptom, Root cause, Consequence). The articulation in the Proposal may be preliminary and will be sharpened for Stage 2.
- 5) References: Use a consistent format (e.g., APA) for all references, including books, academic journals, online articles, blogs, etc.

Stage 2 – Presentation

Deadline: May 17 (Sunday), 09:00–12:00 (Beijing Time)

Deliverable: 20-minute presentation + 5-minute Q&A

Goal: Present Stage 1 insights and a sharp, well-argued Problem–Solution Logic: a precisely articulated information-system problem (not more than three), and a focused AI-driven solution whose mechanism is clearly linked to that problem.

Why this stage matters: The most common weaknesses observed in prior iterations of this project were (a) problem statements that relied on generic buzzwords ("data silo," "information cocoon," "governance concern") without specifying whose problem it was, where it manifested, or why it mattered; and (b) AI solutions described by name only ("NLP-based standardization," "real-time

synchronization engine,” “recommendation algorithm”) without explaining what the AI actually does or how it causally addresses the problem.

To avoid these pitfalls, there are two structured templates for your reference: the Problem Articulation Template and the AI Mechanism Template. Together they form the Problem–Solution Logic that your presentation must convey clearly.

Template 1 — Problem Articulation

Articulate your focal problem along the following five elements. Treat each element as a distinct claim that must be supported with specifics from the company and the MIS you analyzed.

Element	What to articulate
Actor	Identify the specific stakeholder whose problem this is — e.g., a particular user segment, business unit, operations team, or the platform itself. <i>Guiding question: Whose work, decisions, or experience is directly affected?</i>
Context	Specify where in the MIS and the business workflow the problem occurs. Refer to concrete processes, modules, or data flows rather than the organization in general. <i>Guiding question: At which step of which workflow does the problem manifest?</i>
Symptom	Describe the observable manifestation: what goes wrong, in what form, and how frequently or at what scale. <i>Guiding question: What do people actually see, experience, or measure that indicates the problem?</i>
Root cause	Explain the structural reason behind the symptom — typically something about system architecture, data design, process logic, or incentive structure — rather than a restatement of the symptom. <i>Guiding question: Why does this keep happening, even when people try to fix it?</i>
Consequence	Link the problem to a concrete business consequence: lost revenue, degraded decisions, customer churn, compliance risk, strategic misalignment, etc. Quantify where reasonable evidence exists. <i>Guiding question: If this problem is not solved, what specifically gets worse, for whom, and by how much?</i>

Template 2 — AI Mechanism

Articulate your AI-driven solution along the following five elements. The goal is to open the black box: make it possible for a thoughtful reader who is not an AI specialist to understand what the system takes in, what it does, what comes out, and why it will help.

Element	What to articulate
Input data	Specify the data the AI system consumes: sources, format, volume, update frequency, and any preprocessing. Indicate whether these data are already available in the existing MIS or must be newly collected. <i>Guiding question: What does the model see, and where does that data come from?</i>
AI method & justification	Name the AI approach at an appropriate level of specificity (e.g., supervised classification, retrieval-augmented generation, reinforcement learning, graph neural network). Justify why this method fits the problem — rather than a rule-based system, a process redesign, or a different AI method. <i>Guiding question: Why is this AI method the right match for this specific problem?</i>
Processing logic	Describe what the model does to transform inputs into outputs: the core learning or inference task, the decision or prediction the model produces, and the conceptual reasoning behind it. Avoid treating the model as a black box. <i>Guiding question: What is the model learning or inferring, and how does it produce its output?</i>
Output & workflow integration	Specify the output format, who or what receives it, and how it integrates into the existing business process. Clarify whether the system augments human decisions, automates them, or triggers downstream workflows. <i>Guiding question: Where does the output go, and who acts on it?</i>
Causal link to the problem	Map the AI solution back to the problem articulation. Specify which element(s) of the problem (actor, symptom, root cause, consequence) the solution addresses, and through what causal mechanism. <i>Guiding question: Which part of the problem does this solve, and through what mechanism?</i>

Additional Requirement (suggested):

- 1) Recap: Summarize key insights from Stage 1 – core business context, strategic analysis, information system analysis, challenges and problems
- 2) Scope discipline: Propose one focused AI solution directly tied to the focal problem. Do not list broad or multiple unrelated AI ideas.
- 3) Realism: The solution should be technically plausible and appropriately scoped given typical enterprise data, compute, and organizational constraints.
- 4) Expected business value: Link value claims back to specific problem consequences. Do not use fabricated statistics generated by AI.
- 5) Risks and governance: Identify the most salient risks (e.g., data quality, model bias, privacy, change management) and relevant ethical or governance considerations, and briefly note how they would be mitigated.
- 6) Presentation quality: Distribute speaking turns across group members; coordinate transitions; prepare for Q&A on the mechanism, not only on the outcome.

Stage 3 – Final Report

Deadline: May 23 (Saturday), 23:59 (Beijing Time)

Deliverable: A Word document (~15 pages, Times New Roman, 12-point font, 1.5-line spacing) with page numbers, group information, a clear title and well-organized sections.

Goal: Deliver a cohesive and well-structured report that integrates the strategic analysis from Stage 1 and the AI-driven solution from Stage 2 into a clear and actionable proposal.

Report Structure (suggested):

- 1) Executive summary: Concise overview of the focal problem, proposed AI solution, and expected business value.
- 2) Company background: Overview of the company and its core business.
- 3) Strategic and Information System Analysis: Apply key frameworks (e.g., *Five Forces*, *Value Chain*, *Resource-Based View*, *IS Strategy Triangle*) to analyze the business context and the selected MIS.
- 4) Problem definition: Present the focal problem using the Problem Articulation Template. All five elements (Actor, Context, Symptom, Root cause, Consequence) should be developed with supporting evidence.
- 5) Proposed AI-Driven Solution: Present the solution using the AI Mechanism Template (Input data; AI method & justification; Processing logic; Output & workflow integration; Causal link to the problem).
- 6) Expected Business Value: Link claimed value to specific problem consequences. Avoid fabricated statistics generated by AI.
- 7) Risks, Ethical and Governance Considerations: Identify principal risks and describe mitigation.
- 8) Conclusion and Managerial Implications: Summarize the strategic significance of the proposed solution for the company and the broader industry context.
- 9) References: Use a consistent citation format (e.g., APA).

Academic Integrity and AI Use

- All submissions must comply with academic integrity standards. The Turnitin similarity score should be below 15%.
- Clearly disclose whether and how AI tools were used in completing your work (e.g., proposal, presentation, final report).
- AI may be used for drafting support, literature search assistance, and language editing. AI may not be used to fabricate data, invent citations, or generate statistics that are presented as factual evidence.